Raunaq Singh

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Education

Georgia Institute of Technology August 2026 – May 2029
B.S. in Industrial Engineering, AI/OR for Decisions; Minor in Computing & Theory *Atlanta, GA*

University of Illinois Urbana-Champaign August 2025 – May 2026
Grainger College of Engineering (transferred) *Urbana, IL*

Coursework: Data Science Discovery, Data Science Exploration, Calculus I–II, Linear Algebra, Discrete Mathematics

Experience

Software Engineering Intern June 2026 – September 2026
MiraPhoto (early-stage startup) *New York City, NY*

- Shipped a live Organization Profile and RSVP feature by rebuilding its UI from a design screenshot
- Enabled static hosting on GitHub Pages and Vercel by migrating Next.js to Astro with GitHub Actions
- Secured 3 user roles and 4 form inputs by building role-based access control and Zod validation
- Linked the frontend to a live Node/MongoDB API by tracing and fixing an undocumented schema mismatch

Projects

PokéLab | Live App | Code November 2026

- Launched a live app on 1,025 Pokémon by building a cached, validated pipeline over PokéAPI and Smogon
- Found no stable stat archetypes (best ARI 0.59 vs. 0.60 bar) by bootstrap-testing K-Means and GMM
- Beat greedy and 10,000 random teams (4% met all rules) by modeling team picks as an integer program

Mean-Reversion Pairs Trading | Code September 2026 – October 2026

- Removed look-ahead bias and 4 other backtest flaws by adding cointegration tests and a train/test split
- Screened 8 candidate pairs for cointegration using Engle-Granger tests with Benjamini-Hochberg FDR
- Found 1 significant pair (KO-PEP, adj. $p = 0.036$) but no reliable edge after cost and walk-forward tests

Leadership

Co-Founder June 2022 – August 2024
Community Swim *Hillsborough, NJ*

- Taught swimming and water safety to 15 children ages 4–12 by co-founding a summer swim program
- Prepared 10 of 15 students to join a swim team by designing lessons tailored to each child's skill level

Varsity Swimmer; Team Captain (2024–2025) November 2021 – February 2025
Hillsborough High School Swim Team *Hillsborough, NJ*

- Selected team captain for the 2024–2025 season after competing on varsity for four years
- Led and mentored teammates at practices and meets, keeping a supportive team environment

Technical Skills

Languages: Python, TypeScript, JavaScript
Data & ML: pandas, NumPy, scikit-learn, SciPy, statsmodels, Matplotlib, seaborn, Plotly, Jupyter
Optimization & Apps: PuLP (integer programming), Streamlit
Web: Next.js, React, Astro, Tailwind CSS, Zod
Tools: Git, GitHub Actions, pytest, Vercel
Interests: Swimming, Cycling, Fishing, Grilling